Introduction to Computer System

Binary Addition:-

$0+0=0$

$0+1=1$

$1+0=1$

$1+1=10$ → Sum

Carry

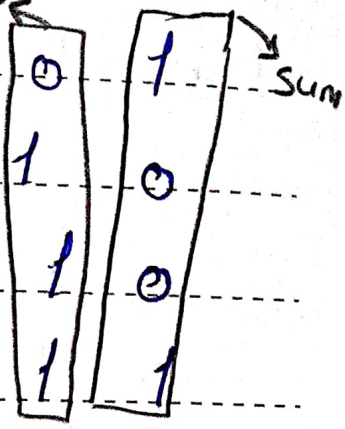
Carry

$1+0+0=01$ → Sum

$1+1+0=10$

$1+0+1=01$

$1+1+1=11$



Ex

① $\begin{array}{r} 1\ 1 \\ 1^{+} \\ \hline 1\ 0\ 0 \end{array} \equiv \begin{array}{r} 3 \\ 1^{+} \\ \hline 4 \end{array}$

② $\begin{array}{r} 1\ 1 \\ 1\ 1^{+} \\ \hline 1\ 1\ 0 \end{array} \equiv \begin{array}{r} 3 \\ 3^{+} \\ \hline 6 \end{array}$

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③
$$\begin{array}{r} 1\,1\,0\,1 \\ +\;1\,0\,1 \\ \hline 1\,0\,1\,0 \end{array}$$

④
$$\begin{array}{r} 1\,0\,0\,0\,1 \\ +\;1\,1\,1\,0\,1 \\ \hline 1\,0\,1\,1\,1\,0 \end{array}$$

⑤
$$\begin{array}{r} 1\,0\,1\,1\,1 \\ +\;1\,1\,0\,0\,0\,1 \\ \hline 1\,0\,0\,1\,0\,0\,0 \end{array}$$

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Binary subtraction

$0 - 0 = 0$

$1 - 1 = 0$

$1 - 0 = 1$

$0 - 1 = 1$ borrow of 1

Ex ①

$$\begin{array}{r} 1\ 1 \\ 0\ 1^{-} \\ \hline 1\ 0 \end{array} \equiv \begin{array}{r} 3 \\ 1^{-} \\ \hline 2 \end{array}$$

②

$$\begin{array}{r} {}^{0}\not{1}\ {}^{2}0\ 1 \\ 0\ 1\ 1^{-} \\ \hline 0\ 1\ 0 \end{array}$$

③

$$\begin{array}{r} 1\ {}^{0}\not{1}\ {}^{2}0\ 1\ 1 \\ 1\ 0\ 1\ 1\ 0^{-} \\ \hline 0\ 0\ 1\ 0\ 1 \end{array} \equiv \begin{array}{r} 27 \\ 22^{-} \\ \hline 5 \end{array}$$

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Binary Multiplication

0 * 0 = 0

0 * 1 = 0

1 * 0 = 0

1 * 1 = 1

Steps

1. Multiply
2. Add

Ex

(1)

$$\begin{array}{r} 11 \\ \times\ 11 \\ \hline 11 \\ 011 \\ + \\ 110 \\ \hline 1001 \end{array} \equiv \begin{array}{r} 3 \\ \times\ 3 \\ \hline 9 \end{array}$$

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(2) 1 1 1

1 0 1 *

1 1

1 0 0 1 1 1

0 0 0 0 0 +

1 1 1 0 0

1 0 0 0 1 1

7

5 *

35

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Binary Divison

① Division
② Multiply
③ Subtraction

Ex 110/11

①
```
   0 1 0
 11 | 1 1 0
      1 1 ↓
   -------
     0 0 0
```

Compare divisor with dividend

① 11 > 1 ?

② 11 > 11 ?

③ 11 > 000 ?

②
```
    0 0 1 1 1
 11 | 1 0 1 0 1
      1 1
   ---------
      1 0 0
        1 1
   ---------
        0 1 1
          1 1
   ---------
            0
```

Ex basic concept

```
    0 1 6
 8 | 1 2 8
     0
   -----
     1 2
     8
   -----
       4 8
       4 8
   -----
         0
```

Ex
```
    0 7 1
 2 | 1 4 2
     0
   -----
       1 4
       1 4
   -----
         0 2
           2
   -----
           0
```

WIN

③

```
      0011
   ┌──────
100 │1100
    100
    ----
     100
     100
     ---
       0
```

④

```
      00101
   ┌──────
101 │11001
    101
    -----
     0010 1
      101
      ---
        0
```

⑤

```
      00011
   ┌──────
1000 │11000
     1000
     ----
      1000
      1000
      ----
         0
```

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